Unit: Unit 2: Exponents Radicals **Name:** ________________
Topic: Exponential Equations **Date:** __________

Quiz (Habanero)

Q1. Solve for x: $2^x = 32$
(A) $x = 3$
(B) $x = 4$
(C) $x = 5$
(D) $x = 6$
(E) $x = 7$

Q2. If $3^{x+1} = 81$, then $x =$
(A) $x = 1$
(B) $x = 2$
(C) $x = 3$
(D) $x = 4$
(E) $x = 5$

Q3. A population of bacteria grows according to the equation $P(t) = 200cdot2^{0.5t}$, where t is time in hours. If $P(t) = 800$, then $t =$
(A) $t = 2$
(B) $t = 3$
(C) $t = 4$
(D) $t = 5$
(E) $t = 6$

Q4. Solve for y: $4^y = 2^{2y}cdot2^3$
(A) $y = 1$
(B) $y = 2$
(C) $y = 3$
(D) $y = 4$
(E) $y = 5$

Q5. The temperature of a cup of coffee decreases according to the equation $T(t) = 95cdot(0.8)^t$, where t is time in minutes. If $T(t) = 65$, then $t =$
(A) $t = 5$
(B) $t = 6$
(C) $t = 7$
(D) $t = 8$
(E) $t = 9$

Q6. Solve for x: $x^2cdot2^x = 64$
(A) $x = 2$
(B) $x = 3$
(C) $x = 4$
(D) $x = 5$
(E) $x = 6$

Q7. The depth of water in a lake decreases according to the equation $D(t) = 50cdot(0.9)^t$, where t is time in days. If $D(t) = 35$, then $t =$
(A) $t = 3$
(B) $t = 4$
(C) $t = 5$
(D) $t = 6$
(E) $t = 7$

Q8. Solve for y: $3^{y+1} = 27cdot3^2$
(A) $y = 1$
(B) $y = 2$
(C) $y = 3$
(D) $y = 4$
(E) $y = 5$

Open Ended Questions (FRQ)

Q9. A hurricane's wind speed decreases according to the equation $W(t) = 120cdot(0.8)^t$, where t is time in hours. Find the time when the wind speed is 80 mph, and show your work.

Q10. The population of a city grows according to the equation $P(t) = 100cdot2^{0.2t}$, where t is time in years. Find the time when the population is 250, and show your work.

Chef's Tips

- Tip 1: Use logarithms to solve exponential equations
- Tip 2: Check your units and dimensions
- Tip 3: Simplify your expressions before solving